

COMP 1111 Final

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I. QUESTION (20 POINTS)

Write a Java method, named **copyN**, that takes an integer array, A , and two integers, n and m , as arguments, where $n, m > 0$. **copyN** copies the each m 'th element of the array A n times and returns the resulting array back.

$A = \{2, 4, 5, 6\}$ $n = 3, m = 2$.

$\{4, 4, 4, 6, 6, 6\}$

$A = \{3, 1, 2\}$ $n = 2, m = 3$.

$\{2, 2\}$

II. QUESTION (20 POINTS)

Write a Java method **counts**, that takes an integer array and counts the occurrences of each member of the array and returns those occurrences as an array. You can assume the elements of the array are non-negative and less than 100.

Write another Java method **maxCount**, that takes an integer array and returns the count of the most frequent member. You must use **counts** method in this second method.

$A = \{6, 2, 4, 5, 6, 2, 3, 6, 8, 4, 6, 8, 6\}$
Returns 5

$A = \{2, 3, 2, 1, 1, 1, 2, 1\}$
Returns 4.

III. QUESTION (20 POINTS)

Write a Java method **sortAlternate**, which takes a 2d array A as argument and sorts the rows of the arrays alternately. First row is sorted in increasing order, second row in decreasing order, third row in increasing order, etc. Write helper methods

sortIncreasingOrder(int[][] A, int rowNo)

sortDecreasingOrder(int[][] A, int rowNo),

which sorts the row of the 2d-array A with index $rowNo$ in increasing and decreasing order respectively. Then use those helper methods in **sortAlternate**.

$$A =$$

3	7	12	1
1	8	4	6
6	14	0	2

$$A' =$$

1	3	7	12
8	6	4	1
0	2	6	14

IV. QUESTION (20 POINTS)

Suppose that we are modelling AmiralBattı game. Write the Java method **int numberOfHits(boolean[][] shots, boolean[][] shipParts)**, which counts the number of ship parts that are hit. In this function; $shots[i][j]$ is true, if we have shot the cell in the i 'th row and j 'th column; false otherwise. Yet in this function; $shipParts[i][j]$ is true, if there is a ship part in the cell in the i 'th row and j 'th column.

$$shots =$$

true	true	false	false
false	false	false	false
true	true	false	false
false	false	false	false

$$shipParts =$$

true	false	true	true
false	false	false	false
false	true	true	true
false	false	false	false

numberOfHits returns 2.

V. QUESTION (20 POINTS)

Write a method **paste**, which takes two arrays A and B and a value k as arguments, and returns a new array which inserts the array B at position k in the array A and returns the resulting array.

$A = \{2, 4, 5, 6\}$, $B = \{8, 10, 12\}$, $k = 2$.

$\{2, 4, 8, 10, 12, 5, 6\}$

$A = \{3, 4, 8\}$, $B = \{8, 10, 12, 20, 24\}$, $k = 0$.

$\{8, 10, 12, 20, 24, 3, 4, 8\}$